

# Hridoy A. Hazarika

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## EDUCATION

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### Tezpur University

*Integrated Msc - Mathematics*

Tezpur, Assam, India

*Aug. 2024 – May 2029*

## EXPERIENCE

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### Undergraduate Research Student

*Tezpur University*

Tezpur, Assam, India

*Aug. 2024 – Present*

- Studied advanced topics in linear algebra, abstract algebra, real analysis, and combinatorics beyond coursework.
- Prepared mathematical documents, assignments, and reports using  $\text{\LaTeX}$ .
- Used MATLAB for computational exploration and visualization of mathematical concepts.

### Summer Research Intern

*Indian Institute of Technology Guwahati*

Guwahati, Assam, India

*June 2026 – Jul. 2026*

- Completed a research internship under the supervision of Dr. Arup Chattopadhyay.
- Studied the geometric foundations of inner product spaces, including orthogonal projections, linear functionals, hyperplanes, and Hilbert spaces.
- Prepared a comprehensive research report in  $\text{\LaTeX}$  with rigorous mathematical exposition and proofs.

### Independent Research Study

*Analytic Number Theory*

Remote

*May 2026 – Present*

- Independently studying analytic number theory with emphasis on  $q$ -series, partition theory, and modular forms.
- Reading research papers and classical texts to build foundations for future research.

## TECHNICAL SKILLS

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**Programming:** C, Python (Basic), MATLAB, SQL

**Typesetting:**  $\text{\LaTeX}$  (Overleaf, mathematical typesetting, TikZ)

**Mathematical Software:** MATLAB, GeoGebra

**Productivity:** Microsoft Excel, Microsoft Word, Microsoft PowerPoint

**Version Control:** Git, GitHub

**Operating Systems:** Windows, Linux (Basic)